

Electronic Support Scan Scheduling

Different papers optimize different parts of the same problem:

which band to observe, when to observe it, how long to dwell, and how to adapt as emitter behaviour becomes known

Clarkson — *Optimal Periodic Sensor Scheduling in Electronic Support*: This is the clearest paper for the **time-allocation decision**. The receiver repeatedly sweeps a fixed set of frequency bands, and the scheduler decides **how much dwell time to allocate to each frequency band** and, in the extended optimization, **what the overall sweep period should be**. The frequency order itself is not the important optimization variable; the paper explicitly says the order can remain fixed as long as it repeats consistently. Its objective is to minimize the **maximum intercept time across all known threat emitters**. So, in this paper, the scheduler decision is essentially **dwell time per band + overall sweep period**. It assumes reliable prior information about emitter RF, scan period, beamwidth and PRI, and assumes periodic behaviour.

Glaude et al. — *Learning of Scanning Strategies for Electronic Support Using Predictive State Representations*: Here the main decision becomes **which frequency band to listen to at the next time step**. The receiver observes its history of frequency choices and interceptions/misses, learns the hidden behaviour of each radar, predicts the probability of a future interception for each radar, and then chooses the next frequency band using an exploration/exploitation strategy. Time is present because the decision is made at every successive time step, so the sequence of chosen frequencies determines when bands are revisited, but the paper does not make dwell duration a separate learned decision. Thus its scheduler decision is mainly **next frequency band**, while **time/revisit behaviour emerges from the sequence of those choices**.

Gul & Erer — *Scanning Strategy Learning for Electronic Support Receivers by Robust Principal Component Analysis*: This paper keeps essentially the **same scheduling decision as the PSR paper**. It still uses Transformed Predictive State Representation to model the emitter/receiver dynamics and then uses a stochastic multi-armed-bandit-style strategy to choose the **next frequency band**. Its main difference is how the underlying predictive model is learned: it replaces the previous Singular Value Thresholding approach with Robust Principal Component Analysis. So this paper does **not introduce a fundamentally new scheduler action**; its novelty is mainly the **better/alternative model-learning method** behind the scheduler.

Apfeld et al. — *An Adaptive Receiver Search Strategy for Electronic Support*: This one is particularly interesting for our problem because it makes the decision more explicitly **frequency + timing related**. Initially the receiver randomly explores frequency bands. Once it detects a band, it gives that band more attention and observes its SNR over time. It then uses autocorrelation to estimate the radar's scan period and, once that estimate becomes stable,

schedules future receiver dwells around the predicted times of maximum SNR. Therefore, its scheduler effectively decides **which bands deserve attention and when those bands should be visited**, using learned timing information. It can also increase the number of dwells around an expected peak when the prediction becomes unreliable and restart exploration after repeated misses.

Teissier et al. — *Interception Model of Random Scanning Strategies Against Frequency-Agile Radar in Electronic Support* (2026): This paper makes the **time dimension explicit**, but in a different way. It models a randomly scanning receiver against frequency-agile and spatially scanning radar and derives formulas for Probability of Intercept, Intercept Time and Pulse Interception Ratio. Its optimization then uses a minimax approach to find the **optimal dwell time** for multiple known emitters while considering uncertainty. Importantly, the paper assumes the random frequency-selection mechanism and focuses on optimizing receiver parameters such as dwell time; it explicitly says its random scan uses **constant dwell time** during an observation sequence. So its primary decision variable is **dwell time**, not an adaptive next-frequency decision based on learning.

Source	Primary decision	Prior information	Time explicit?	Online adaptation	Our use
Clarkson 2005	Dwell time per band + sweep period	High	Yes	No	Classical time baseline
Glaude 2015	Next frequency band	Learned from observations	Indirect via time steps	Yes	Learning baseline
Gul & Erer 2021	Next frequency band	Learned from observations	Indirect via time steps	Yes	PSR/TPSR + RPCA baseline
Apfeld et al.	Band focus + revisit timing	Starts with little knowledge	Yes	Yes	Adaptive timing baseline
Teissier 2026	Dwell-time optimization for random scan	Known emitter parameters	Yes	No	Analytic IT/POI/PIR baseline
McWhorter 2021	EW countermeasure / threat action	Detected/classified radar info	Mission-cycle time	Partly	Prediction-decision reference

Existing research has explored frequency selection, adaptive revisit timing, and dwell-time optimization separately or in limited combinations. Apfeld already demonstrates adaptive frequency focus and timing based on learned emitter behaviour. Our research opportunity is therefore not simply to combine frequency and time, but to investigate a unified closed-loop scheduler that can dynamically allocate limited sensing opportunities among multiple uncertain emitters while explicitly optimizing interception probability and intercept time